

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL3)*

Assessment Tool Weightage: 40%

Dr A.Manimaran

QUESTIONS:

Total Marks:40

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Ragul

Annexure A

TECHNICAL PRESENTATION

1. Selecting an Appropriate SDLC Model for Mobile App Development

A company develops a Mobile Learning Application where customer requirements change frequently.

Compare Agile, Waterfall, and Spiral models. Analyze their advantages and limitations and recommend the most suitable SDLC model with justification.



SIMATS
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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Selecting an Appropriate SDLC Model for Mobile App Development

CSA-1108 (OOAD)

Incharge :
Dr.A.Manimaran

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INTRODUCTION



- SDLC (Software Development Life Cycle) is a structured process for developing software.
- It ensures high -quality software with proper planning, design, testing, and maintenance.
- Different SDLC models are selected based on project size, complexity, and customer requirements.
- Mobile learning applications require continuous updates and user feedback.



Waterfall Model



Definition:

The Waterfall model is a sequential software development approach where each phase is completed before moving to the next.

Advantages:

- Simple and easy to understand.
- Well-defined documentation.

Limitations:

- Difficult to accommodate changing requirements.
- High cost of modifications.

Best Used For:

- Banking systems
- Government projects
- Small projects with stable requirements



Agile Model

**Definition:**

Agile is an iterative and incremental development model that delivers software in small, frequent releases.

Advantages:

- Improved software quality.
- Quick bug fixing.

Limitations:

- Requires experienced developers.
- Less documentation.
- Difficult cost estimation.
- Frequent customer participation is necessary.

Best Used For:

- Mobile applications
- Startups



Spiral Model

**Definition:**

The Spiral model combines iterative development with risk analysis.

Advantages:

- Excellent risk management.
- Suitable for large projects.
- Continuous customer feedback.
- Flexible development process.

Limitations:

- Complex to manage.
- Requires risk management expertise.

Best Used For:

- Aerospace
- Banking software



Comparison of SDLC Models



 Feature	 Waterfall Model	 Agile Model	 Spiral Model
 Requirement Changes	Poor	Excellent	Good
 Customer Involvement	Low	High	High
 Flexibility	Low	High	High
 Development Cost	Low	Medium	High
 Risk Handling	Low	Medium	Excellent
 Delivery Speed	Slow	Fast	Medium
 Documentation	High	Medium	High
 Best Suited For	Fixed and Well-Defined Requirements	Frequently Changing Requirements	Large, Complex, and High-Risk Projects



Recommended SDLC Model



- **Recommendation** : Agile Model

Justification :

- Customer requirements change frequently.
- New features can be added after every iteration.
- Continuous user feedback improves application quality.
- Faster release cycles reduce development time.
- Easy bug fixing and feature enhancement.
- Higher customer satisfaction.
- Supports continuous testing and integration.

Conclusion : Agile provides flexibility, faster development, and better user satisfaction compared to Waterfall and Spiral models.



Conclusion & References



Conclusion:

- Selecting the correct SDLC model improves software quality and project success.
- Waterfall is suitable for projects with fixed requirements.
- Spiral is ideal for high-risk and complex projects.
- Agile is the best choice for mobile learning applications.

References:

- Roger S. Pressman – Software Engineering: A Practitioner's Approach
- Ian Sommerville – Software Engineering
- IEEE Xplore Digital Library
- ACM Digital Library
- Agile Manifesto (2001)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or	Poor organization, unclear speech,

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		delivery		confidence	ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness